THE ISOM FUNCTOR IS AN FPQC SHEAF

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-hom (canonical tag 08JT). cited at statement line 993. Standing situation.
- Proposition spaces-descent-proposition-fpqc-descent-quasi-coherent (canonical tag 04W8). cited at proof line 1009. Descent of quasi-coherent modules along fpqc coverings.

Source reference. Original definition 08JT: sources/stacks-chow/tags.txt, source lines 10861–10861. Cited at statement line 993: the standing situation of the Hom/Isom section.

Source reference. Original lemma 03MR: sources/stacks-chow/tags.txt, source lines 4759–4759. Cited at statement line 1006: the covering {X_i -> X_T} is an fpqc covering of X_T.

Source reference. Original lemma 04W8: sources/stacks-chow/tags.txt, source lines 6351–6351. Cited at statement line 1009: fpqc descent of quasi-coherent modules gives that u is an isomorphism.

Lemma 0.2. *In Situation 08JT the functor $Isom(\mathcal{F}, \mathcal{G})$ satisfies the sheaf property for the fpqc topology.*

Proof. We have already seen that $Hom(\mathcal{F}, \mathcal{G})$ satisfies the sheaf property. Hence it remains to show the following: Given an fpqc covering $\{T_i \to T\}_{i \in I}$ of schemes over B and an $\mathcal{O}_{X_T}$-linear map $u : \mathcal{F}_T \to \mathcal{G}_T$ such that u_{T_i} is an isomorphism for all i, then u is an isomorphism. Since $\{X_i \to X_T\}_{i \in I}$ is an fpqc covering of X_T, see Topologies on Spaces, Lemma 03MR, this follows from Descent on Spaces, Proposition 04W8. □

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